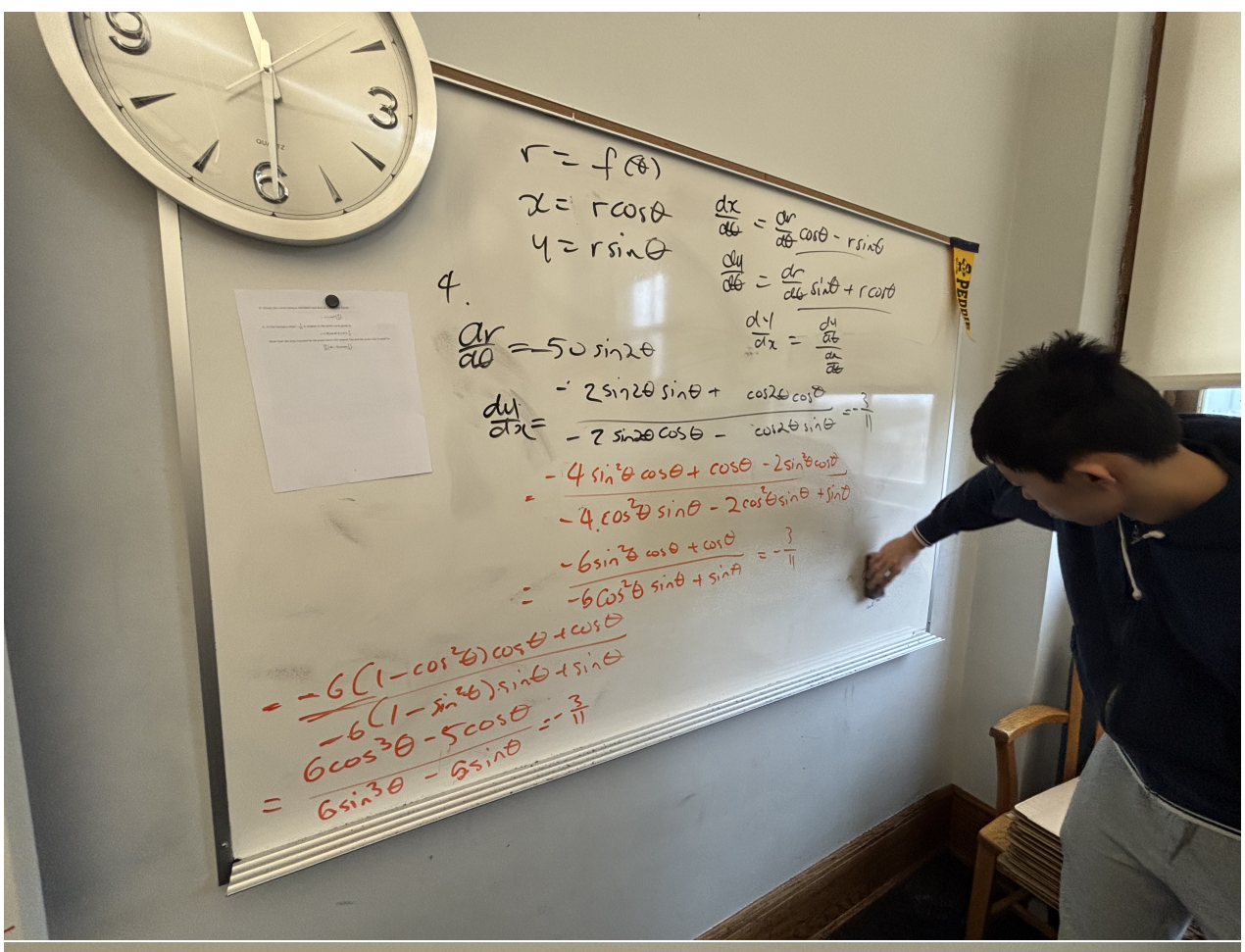
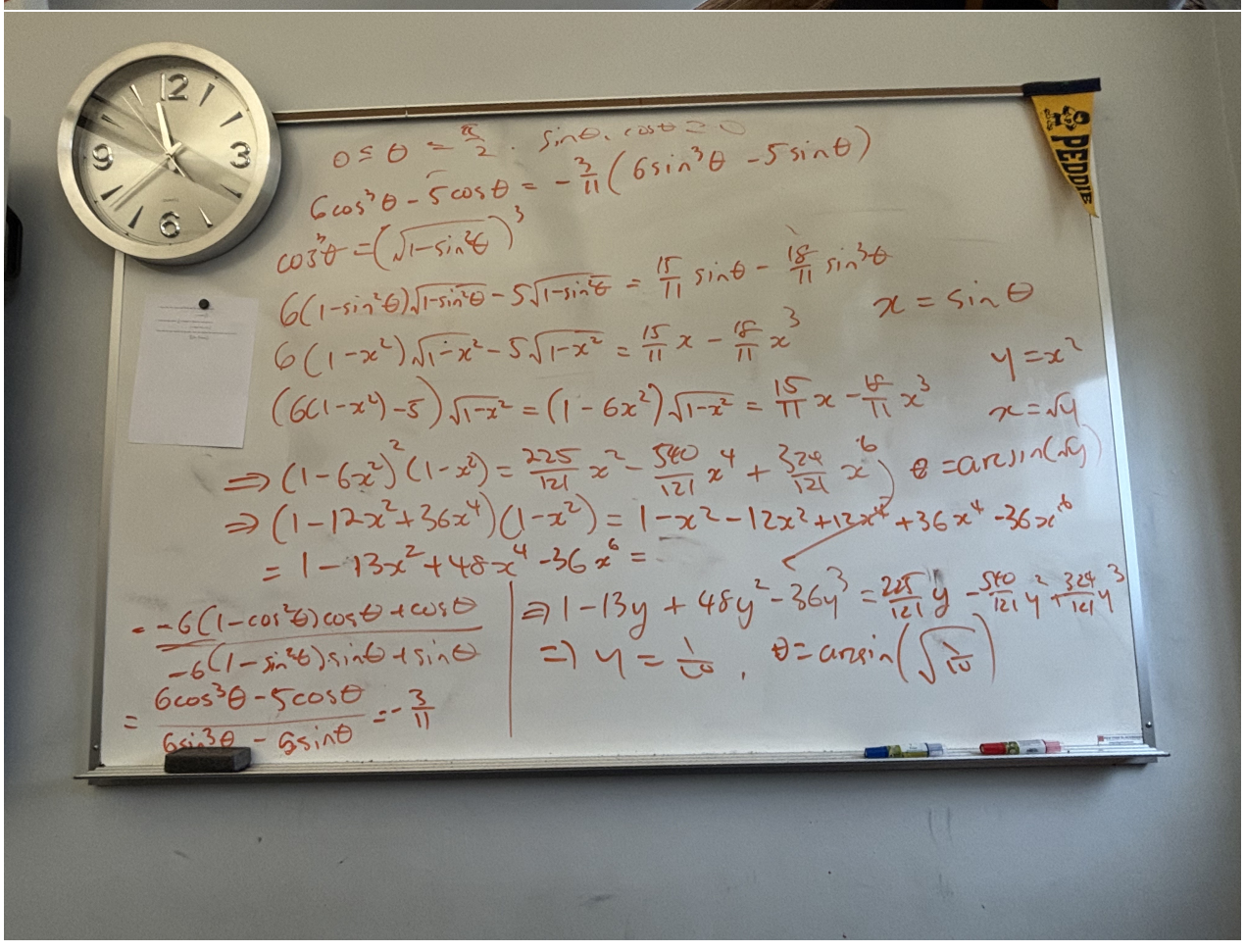
Problem Set #62

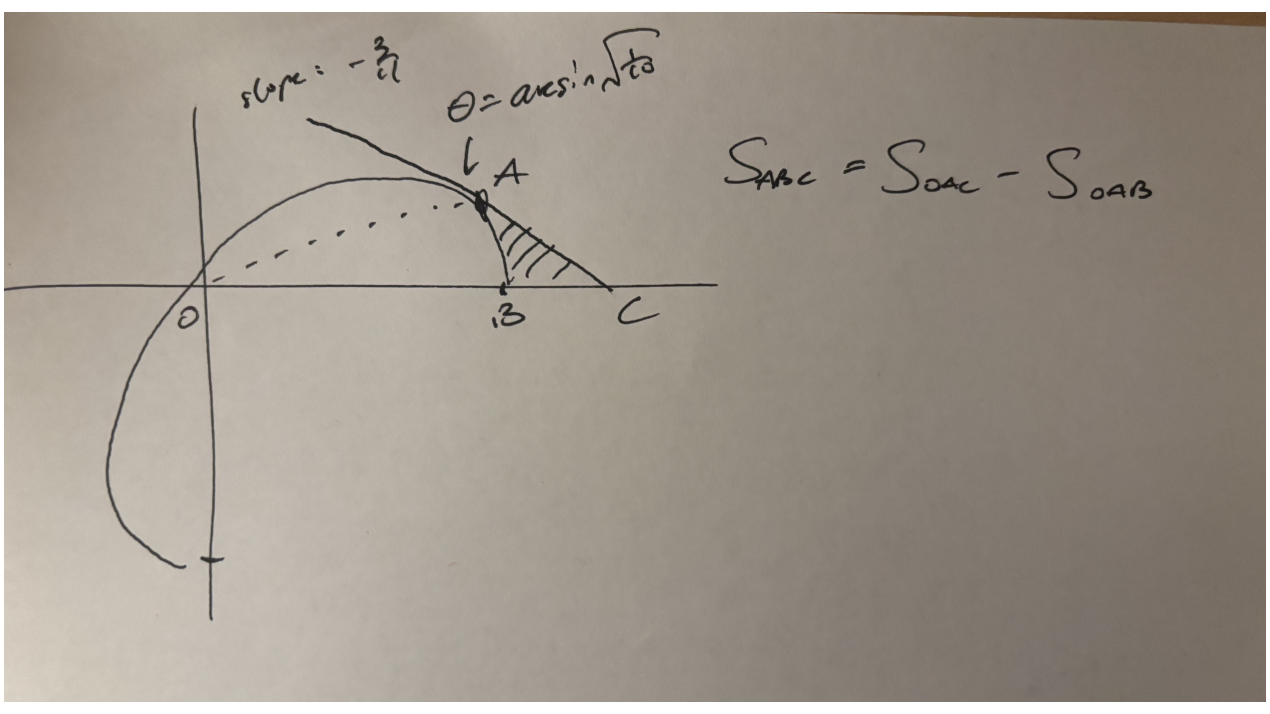
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Problem 4







$$r = f(\theta) = 25\cos 2\theta$$

It is known that the slope of the tangent line at $\theta = \varphi = \arcsin\left(\sqrt{1/10}\right)$ is $-3/11$. Then, we know:

$$\sin\varphi = \sqrt{\frac{1}{10}} = \frac{1}{\sqrt{10}}$$

$$\cos\varphi = \sqrt{1 - \left(\sqrt{\frac{1}{10}}\right)^2} = \sqrt{1 - \frac{1}{10}} = \sqrt{\frac{9}{10}} = \frac{3}{\sqrt{10}}$$

$$\cos 2\varphi = 1 - 2\sin^2\varphi = 1 - 2\left(\sin\varphi\right)^2 = 1 - 2\cdot\frac{1}{10} = \frac{5}{5} - \frac{1}{5} = \frac{4}{5}$$

$$\sin 2\varphi = 2\sin\varphi\cos\varphi = 2\cdot\frac{1}{\sqrt{10}}\cdot\frac{3}{\sqrt{10}} = \frac{6}{10} = \frac{3}{5}$$

The coordinates of each point O, A, B, C are known:

$$\begin{aligned} O &= (0,0) \\ A &= (r\cos\varphi, r\sin\varphi) = (25\cos 2\varphi\cos\varphi, 25\cos 2\varphi\sin\varphi) \\ &= \left(25\cdot\frac{4}{5}\cdot\frac{3}{\sqrt{10}}, 25\cdot\frac{4}{5}\cdot\frac{1}{\sqrt{10}}\right) = \left(\frac{60}{\sqrt{10}}, \frac{20}{\sqrt{10}}\right) \\ B &= (25\cos 0\cos 0, 25\sin 0\cos 0) = (25, 0) \end{aligned}$$

From here, we can find the tangent line whose slope is $-3/11$, and passes through point A.

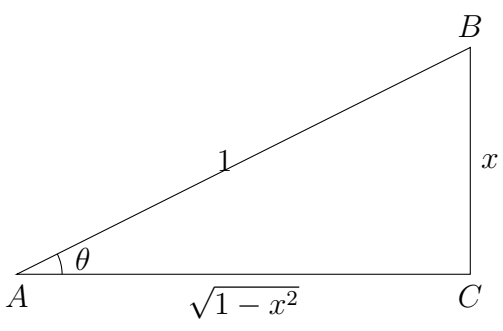
$$\begin{aligned} y - y_0 &= m(x - x_0) \\ y - \frac{20}{\sqrt{10}} &= -\frac{3}{11}\left(x - \frac{60}{\sqrt{10}}\right) \end{aligned}$$

To find C, we need to calculate its polar-axis-intercept, or x-intercept:

$$\begin{aligned} -\frac{20}{\sqrt{10}}\cdot\left(-\frac{11}{3}\right) &= x - \frac{60}{\sqrt{10}} \\ x &= \frac{220}{3\sqrt{10}} + \frac{60}{\sqrt{10}} \end{aligned}$$

Thus, we have $C = \left(\dfrac{220}{3\sqrt{10}} + \dfrac{60}{\sqrt{10}}, 0\right)$. Knowing the altitude and base of triangle $\triangle OAC$, we can find its area. We can also find the area bound by the polar curve and the polar axis by integration.

First, let us find how to represent the arcsine of a number as the arctaengent of another number. Let $\theta = \arcsin x$, then $\sin\theta = x$



$$46\tan\theta = \frac{x}{\sqrt{1-x^2}} \implies \theta = \arctan\left(\frac{x}{\sqrt{1-x^2}}\right)$$

$$\begin{aligned} S_{OAC} &= \frac{1}{2}\left(\frac{220}{3\sqrt{10}} + \frac{60}{\sqrt{10}}\right)\left(\frac{20}{\sqrt{10}}\right) = \frac{1}{2}\left(\frac{4400}{30} + \frac{1200}{10}\right) = \frac{220}{3} + \frac{180}{3} = \frac{400}{3} \\ S_{OAB} &= \frac{1}{2}\int_0^{\varphi}\left(f(\theta)\right)^2\,\mathrm{d}\theta = \frac{1}{2}\int_0^{\varphi}\left(25\cos 2\theta\right)^2\,\mathrm{d}\theta = \frac{625}{2}\int_0^{\varphi}\cos^2 2\theta\,\mathrm{d}\theta = \frac{625}{2}\int_0^{\varphi}\frac{1+\cos 4\theta}{2}\,\mathrm{d}\theta \\ &= \frac{625}{4}\left[\theta + \frac{\sin 4\theta}{4}\right]_0^{\varphi} = \frac{625}{4}\left(\varphi + \frac{1}{4}\sin 4\varphi\right) = \frac{625}{4}\left(\varphi + \frac{1}{2}\sin 2\varphi\cos 2\varphi\right) \\ &= \frac{625}{4}\left(\varphi + \frac{1}{2}\cdot\frac{4}{5}\cdot\frac{3}{5}\right) = \frac{625}{4}\left(\arcsin\sqrt{\frac{1}{10}} + \frac{6}{25}\right) = \frac{625}{4}\left(\arctan\left(\frac{\sqrt{\frac{1}{10}}}{\sqrt{1-\frac{1}{10}}}\right) + \frac{6}{25}\right) \\ &= \frac{625}{4}\left(\arctan\left(\frac{\frac{1}{\sqrt{10}}}{\frac{3}{\sqrt{10}}}\right) + \frac{6}{25}\right) = \frac{625}{4}\left(\arctan\left(\frac{1}{3}\right) + \frac{6}{25}\right) \end{aligned}$$

Then we can subtract these two areas, as in the diagram, to obtain the desired area.

$$\begin{aligned} S_{ABC} &= S_{OAC} - S_{OAB} = \frac{400}{3} - \frac{625}{4}\left(\arctan\left(\frac{1}{3}\right) + \frac{6}{25}\right) = \frac{400}{3} - \frac{625}{4}\arctan\frac{1}{3} - \frac{25\cdot 3}{2} \\ &= \frac{800}{6} - \frac{225}{6} - \frac{625}{4}\arctan\frac{1}{3} = \frac{575}{6} - \frac{625}{4}\arctan\frac{1}{3} = \frac{25}{12}\cdot 46 - \frac{25}{12}\cdot 75\arctan\frac{1}{3} \\ &= \boxed{\frac{25}{12}\left(46 - 75\arctan\frac{1}{3}\right)} \end{aligned}$$